

Supplementary Information

Host-Conditioned Spatial Structure of Canopy Mortality in Finland’s Boreal Forests

immediate

This Supplement records the software environment and random seeds (§1), the shape-feature correlation structure (§2) and univariate feature AUCs (§3) that underpin the morphological-gradient framing, and the per-figure provenance trail (§4). It accompanies the main text and uses the same cluster table (14,582 clusters; 3,422 elongated/dispersed “Type 0” and 11,160 compact/aggregated “Type 1”).

1 Software environment and random seeds

Component	Version / value
Python	3.9.6
numpy / scipy	2.0.2 / 1.13.1
pandas / scikit-learn	2.3.3 / 1.6.1
geopandas / shapely / pyproj	1.0.1 / 2.0.7 / 3.6.1
rasterio / pointpats	1.4.3 / 2.3.0
matplotlib	3.9.4
R	4.6.1
spatstat (.geom/.explore)	3.6.1 (3.8.1)
Covariance-preserving null seed	42
Discreteness tests (GMM/PCA) seed	42
Tree-level subsample seed	42
spatstat envelopes	39 sims/block
DBSCAN <code>eps</code> (clusters / blocks)	20 m / 15 km
DBSCAN <code>min_samples</code>	1 (deterministic)
Host/species raster vintage	MS-NFI <code>vm1x_1721</code> (2021)

Table 1: Software versions and seeds for full reproducibility. All stochastic steps are seeded; DBSCAN is deterministic at `min_samples = 1`.

2 Shape-feature correlation structure

The morphological features are strongly intercorrelated (Figure 1). Aspect ratio and elongation are effectively the same axis ($\rho = 1.00$); compactness is its near-perfect inverse ($\rho = -0.85$); area, perimeter, core:edge ratio and point count form a tight size block ($\rho \geq 0.71$). The three features used to define the $k = 2$ partition (fractal dimension, Clark–Evans, aspect ratio; red boxes) therefore span only two largely independent directions—an elongation axis and a size/density axis—which is why a single principal component captures 60% of the variance (main text, §Morphological Gradient) and why the partition is best read as a cut through a continuum rather than two natural classes.

3 Univariate feature separability

Table 2 reports the area under the ROC curve for each feature taken alone as a classifier of the two morphological endpoints (direction: “+” = larger values associate with the compact endpoint). The high AUCs of compactness (0.961), aspect ratio (0.945) and elongation (0.945) are *expected and partly circular*: the partition is defined on aspect ratio and fractal dimension, so any feature coupled to those axes separates the endpoints almost perfectly by construction. These values quantify the geometric redundancy among descriptors discussed in the main text; they are

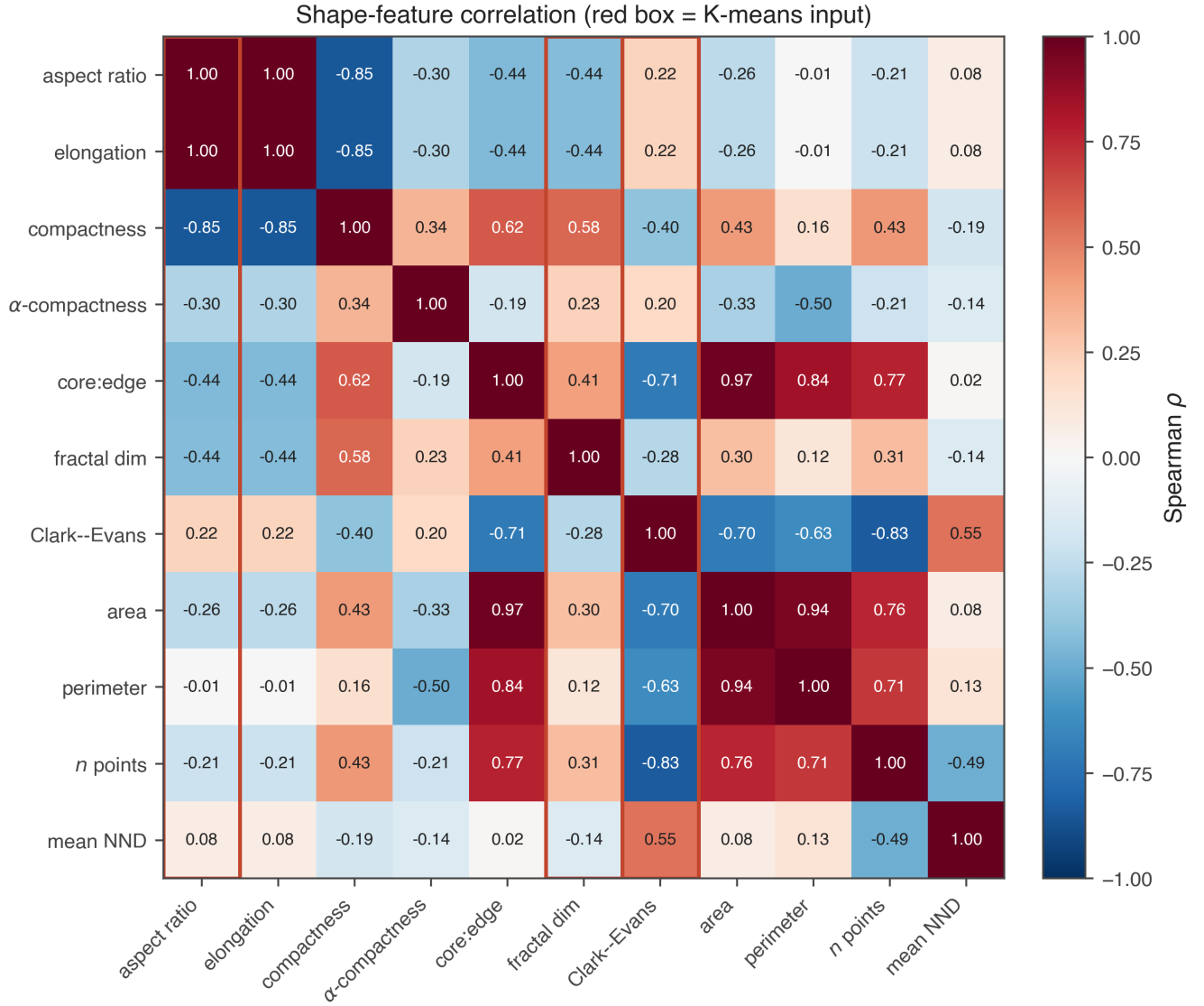


Figure 1: Spearman correlation among the eleven cluster-level shape and point-pattern features. Red boxes mark the three features input to $k = 2$ K-means.

not independent evidence for two classes. The one shape-orthogonal quantity examined in the main text—landscape connectivity—adds no separating signal once size is accounted for.

4 Per-figure provenance

Every main-text figure is regenerated by a script in the public repository (https://github.com/aniskhan25/TreeMortSpatial_analysis/); the `restyle_*` scripts apply the shared house style (`analysis/figstyle.py`) to notebook-produced figures.

Feature	AUC	Dir. (compact)
compactness	0.961	+
aspect ratio [†]	0.945	−
elongation	0.945	−
fractal dim [†]	0.938	+
core:edge	0.837	+
Clark–Evans [†]	0.760	−
area	0.733	+
n points	0.726	+
α -compactness	0.662	+
mean NND	0.622	−
perimeter	0.577	+

Table 2: Univariate ROC-AUC for separating the elongated (Type0) from the compact (Type1) endpoint. [†] = feature used to define the $k = 2$ partition. Direction “+” = larger values associate with the compact endpoint. High AUCs reflect geometric redundancy, not independent validation.

Figure	Generating script	Primary input
fig1_spatial_extent	restyle_maps.py	cluster table
fig2_histograms	make_histograms_figure.py	cluster table
fig3_correlation	restyle_charts.py	cluster table
fig4_scatters	restyle_charts.py	cluster table
fig6_typology_map	restyle_maps.py	cluster table
fig8_local_morans	restyle_maps.py	gate_moran*
fig_shape_space	restyle_charts.py	cluster table
fig_size_dist	restyle_charts.py	cluster table
fig_orientation_rose	restyle_charts.py	cluster table
fig_moran_heatmap	restyle_charts.py	gate_moran*
fig_silhouette_permtest	make_permtest_figure.py	discreteness_tests.py
fig_dbscan_sensitivity	make_gate_figure.py	gate1_eps_stability
fig_anisotropy	make_anisotropy_figure.py	gate3_anisotropy
fig_shap_heldout	restyle_shap.py	held-out RF
fig_contagion_a/b	restyle_maps.py	cluster table
fig_sampling_extent	scope_experiments.py	cluster table
fig_host_conditioned	make_hostcond_figure.py	host_pointprocess.py
host_conditioned_spatstat	host_pointprocess_spatstat.R	per-block grids
fig_hostcond_rigor	make_hostcond_rigor_figure.py	linhom*.csv
fig_species_validation	species_validation.py	MS-NFI species rasters
fig_si_corr (this SI)	make_si_tables.py	cluster table

Table 3: Per-figure provenance. “cluster table” = data/clusters_typed.csv; gate/host artifacts are written to analysis/out/.